Input

2241×1500 image
24 detail tiles



"What is the color of the helmet?"

budget B=1536
K=6 tiles = 2 coverage + 4 detail

① Coverage tiles → prior



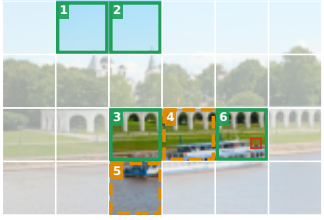
2 coverage tiles: whole image at low resolution, always kept

one partial LLM pass on $[x_b; V^c; x_a]$: attention from question x_a, layers 21–28



prior π_t per detail tile (darker = higher)

② Select detail tiles before the ViT



kept directly: prior top $k-a$ = 2

boundary window: S = 4, scored by q_t

... of which top a = 2 are kept

discarded: 18 tiles, never encoded

numbers on tiles: prior rank

③ Stream into the KV cache



selected detail tiles, near-native resolution
(* = chosen by boundary scoring)



KV cache: 6 tiles × 256 = 1,536 visual tokens
(detail tiles in raster order, then coverage tiles)

ViT encodes 8 of 26 tiles;
no cost term grows with N

Answer: "red" (correct)